

WASTE MANAGEMENT ACTIVITIES UTILIZED IN RESOLVING ENVIRONMENTAL
ISSUES FOR SUSTAINABLE COMMUNITY DEVELOPMENT IN SOUTH EAST STATES,
NIGERIA.

Michael Emeka Ayogu

Department of continuing education and development studies, faculty of education, Enugu State
University of Science and Technology, Agbani, Enugu State, Nigeria

Corresponding Author: mchlayogu@gmail.com,

Abstract

The study centered on determining the extent to which waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria. The study was carried out in South East States, Nigeria using a descriptive survey research design. One research question and one null hypotheses guided the study. The population for the study was 7,536 male and female members of community project management committee in the 942 communities in South East States, Nigeria. The sample size of the study was 420 respondents which comprised 297 male and 123 female members of community project management committee from the communities under study. The instrument used for data collection was a 10-item structured questionnaire which was face validated by three experts; two from the Department of Continuing Education and Development Studies, and one from the Department of Measurement and Evaluation. The reliability of the instrument was determined using Cronbach Alpha and the reliability coefficient of the instrument yielded alpha value of 0.79. The data collected were analyzed using mean and standard deviation while null hypotheses was tested at 0.05 level of significant and appropriate degree of freedom with t-test statistics. The result revealed that the members of community believed that waste management activities were utilized to a high extent in resolving environmental issues for sustainable community development in South East States, Nigeria. The null hypotheses tested showed no significant difference in the mean ratings of male and female members of community project management committee on the extent waste management activities were identified to be utilized in resolving environmental issues for sustainable community development in South East States, Nigeria. Based on the findings, the community project management committee should ensure proper waste management activities in the target communities as a method of resolving environmental issues in their various communities. Also, the government should encourage and support community project management committee to embark on more sustainable community development activities for the good of members of the community and the development of their communities.

Keywords: Waste management activities, environmental issues, community development and sustainable community development.

INTRODUCTION.

The complexity to improve environmental situation threatens every nation in the world. The physical, social, economic, and organizational conditions often vary widely within a country, and certainly among countries. From the global point of view, environmental issues and challenges have for ages, been sprouting on daily basis. Thus, we experience an environment that is either unusually drought, heating up through global warming, burning down from wildfires, being washed away by flooding which translate to gully erosions, or being blown away by high winds typhoons, tornadoes, hurricanes and windstorms. Such occurrence bring about the freezing up, melting down, and inundated with wastes, noise pollution, plastics, toxic materials, household wastes, oil spills, gas flaring, green house and fluorinated gases, electronic wastes, hospital wastes, et cetera. However, fabulous efforts have been made by environmentalists, scientists and cooperate bodies to address and resolve these environmental hazards. As a result, lots of funds are employed into the waste management activities. Even though, some of these efforts have been rewarded with some significant positive results, some environmental officers are completely unacquainted with the effectiveness of these activities in ameliorating the sufferings of community members. Records in various Local Government Areas showed that there are many completed and commissioned waste management activities in various communities for sustainable community development. There are also cases of uncompleted and/or jumbled protection activities within the area that the communities are not interested in. Also, lots of investments put into these waste management activities are not commensurate with their outcome while some are outside the immediate needs of

community members. This is apparent in environmental turbulence which are caused by pollution, floods, litters, poor housing conditions, bush burning, indiscriminate cutting down of trees without replacement, et cetera. These problems gave rise to worrying outcomes in the human society in the form of climate change, diseases, human casualties, environmental hazards and disasters, depletion of the ozone layer, loss of biodiversity and extinction of species, emergence of invasive species as well as pervasive orientations, exploitations, immorality, mass migrations, conflicts and wars, insecurity and mutual distrust between peoples and nations (Poulopoulos, 2016).

This environmental challenge has been recognized, and governments have begun placing restraints on activities that cause environmental degradation. During the Progressive Era in the early 20th century, reformers warned that unregulated economic development was destroying natural resources and raised alarms about the public health crisis of crowded cities, where raw sewage and industrial run-off filled the waterways and smokestack pollution clouded the air that people breathed (Hannah, 2018). Scientific experts, urban reformers, and women groups promoted policies to reduce disease and clean up the air, land, and water. Conservation groups also began mobilizing in the late 1800s to protect wilderness areas and wildlife and regulate logging, mining, dam construction, and other assaults on natural resources (Hannah, 2018). As the environmental philosophy spread, so did the concept of "ecology," an awareness of not only how the natural environment affects human life but also how the activities of humans negatively shape the environment. This worrisome environmental situation challenges human life and community development in different ways and thus, requires different strategies

to ensure a sustainable community development. One of the strategies required to address environmental challenge is through waste management activities.

Waste management activity is an issue that should not be separated from the issue of human activities and health. When the environment is properly managed and protected in terms of adequate environmental governance, promotion of environmental social justice, preservation of a clean and healthy environment; then people's health and wellbeing will be protected and they will have a peaceful environment to flourish. A clean and healthy environment in which the water sources are free from poisons and toxins, every form of human and domestic waste is properly disposed, the atmosphere is free from poisonous gasses is a booster to human health. Thus, the main goal of the National Policy on the Environment (2016) is to achieve sustainable environment in Nigeria and in particular to:

- a. Secure quality of environment adequate for good health and well being.
- b. Conserve and use the environment and natural resources for the benefit of future generations.
- c. Restore, maintain and enhance the eco-systems and ecological processes essential for functioning of the biosphere to preserve biological diversity and the principles of optimum sustainable yield in the use of living natural resources and eco-systems.
- d. Raise public awareness and promote understanding of the essential linkages between the environment, resources and development and encourage individuals and community to participate in environmental improvement efforts.

- e. Co-operate in good faith with other countries, international organization, agencies to achieve optimal use of transbounding natural resources and effective prevention or abatement of transbounding environmental degradation.

There cannot be a hale and hearty nation without a healthy environment. Scientists have warned over the years that an unhealthy population and a degraded environment will hinder progress towards development goals (Torres, 2016). In Nigeria, there is a greater understanding of the principal threats to the environment and public health. These are associated with a variety of factors, including those related to poor Waste Management. There is need to have solemn waste management campaign to help people realize the link between environmental protection, human health, community development, enhanced standard of living and sustainable development.

Community development encompasses all forms of developmental activities in a community. It is an unbounded domain. Cabaj and Weaver (2016) opined that, it seeks to empower individuals and groups of people with the skills they need to effect change within their communities. These skills are often created by formulating social groups that can work for a common agenda. It is also aimed at securing the active participation and co-operation of the people in every community programme that is designed to raise their standard of living and promote their development in every form in order to ensure sustainable community development.

Sustainable community development refers to the process of creating and maintaining thriving, equitable, and environmentally-friendly communities (Harrington, 2016). It involves integrating social, economic, and

environmental considerations to meet the needs of the present without compromising the ability of future generations to meet their own needs (Berg, 2017). From the above, one can see that sustainable community development is crucial for ensuring long-term growth and prosperity.

In Nigeria, environmental management has remained consistently awful over the years and the problem has become a serious issue that has become difficult if not impossible to resolve irrespective of all efforts from all arms of the government. Cities are often over flooded and people made homeless, displaced, and even some die as a result of blocked drainages and waterways filled with waste. Properties and businesses worth millions of naira are often lost in flood arising from blocked drainages. More still, the population has risen rapidly to about 200 million according to the latest National Census of 2006 without commensurate provision of infrastructure and supporting services to cope with the rising population (Holly and Mberu, 2015). The implication here is a wearisome environmental situation noticeable in the form of increase in waste generation, over utilization of limited existing facilities, indiscriminate felling of trees, erosion, lack of awareness, inadequate supply of portable water, poor land use, conversion of residential accommodation to commercial uses, increase in urban slums and shanty shelters of the city as a result of acute rural-urban migration and a general decline in the quality of human life (Okanlawon, Odunjo and Soyebo, 2024). Consequently, the environment continues to deteriorate whereas environmental related diseases aggravate poverty (Torres, 2016). As a result, valuable financial resources that could be invested into meaningful life projects are spent treating sickness and diseases arising from an unclean and unprotected/harmful environment.

This irritating environmental situation has called the attention of different administrations in States of South East Nigeria towards achieving a cleaner and protected environment. Sequel to the fact that these States are densely populated by commercial hub where great amount of industrial trash, solid waste, scrap metals, et cetera abound, the States government have created a long term functional strategy to evacuate, manage, recycle and maintain a feasible incorporated solid waste management system that will efficiently checkmate death-prone environmental hazards (Akinloye, 2018). As a result, the governments of South East States have procured modern solid waste handling evacuation equipment including motorized bins, waste bags, plastic receptacles and many other refuse evacuation materials to ensure proper disposal of refuse in the communities of these States. In addition, the governments introduced environmental activities such as; environmental awareness creation activities, environmental sanitation activities, waste management activities, erosion/flood control activities, tree planting activities, dropping of waste bins in communities and business areas and have purchased new coaster vehicles to convey roads and streets sweepers from one place to the other (Okafor, 2019).

Irrespective of the tireless efforts of the State governments to ensure a cleaner and protected environment in South East States, the attitude of community members in those States towards environmental protection appears to be very poor. Indiscriminate dumping of refuse is still practiced. People dump wastes on gutters especially during rain fall. Such wastes will be carried from one place to another by the water only to block the next flood channel, spread on the road and create unnecessary environmental hazard. The failure of the community members to develop positive attitudes

towards waste management activities could be attributed to some gaps that need to be filled through implementation of waste management programmes. Waste management programmes are germane for the South East States today in taking wise decisions for their own well-being and for the good of the planet and its inhabitants. It therefore becomes imperative for waste management activities to prepare all with critical thinking skills and intellectual framework to face the most critical environmental issues of the century. The task of keeping the earth's surface safe is clearly a joint responsibility of everybody, students must be prepared for those responsibilities as well (Sagdiç and Şahin, 2016).

The environment is a fundamental element of the knowledge base, we need to live in a safe and flourishing World. This calls for urgent implementation of waste management activities in order to protect the environment and ensure healthy living in the communities in South East States, Nigeria.

Waste management could be explained as the supervised handling of waste materials from generation at the source through the recovery processes to disposal. Carrington (2018) while quoting the opinion of United Nations Environmental Protection Agency (UNEPA) maintained that, waste management satisfies the growing demand for emerging scientifically-based information people should have, to avoid degradation of environment. Hence, waste management refers to how to generate, control, store, collect, transport, process and dispose of solid wastes in a manner that is in accordance with the best principles of public health (Budge., Argaw., Jamie., Brown., & Hutchings, 2022). From the above, one can infer that when wastes are managed very well by the community members, it will bring about environmental protection for sustainable community development which

this study is meant to achieve. However, the researcher wishes to study; waste management activities for sustainable community development in South East States, Nigeria.

Statement of the Problem

In the past few years, there have been frequent unparalleled experiences and reports of worrisome environmental issues that adversely affect the integrity of the global environment. We experience an environment that is either unusually drying up as a result of drought, heating up through global warming, burning down from wildfires, being washed away by flooding which translate to gully erosions, being blown away by high winds typhoons, tornadoes, hurricanes, windstorms, freezing up, melting down, inundated with wastes, highly polluted with noise, plastics, toxic materials, household wastes, oil spills, gas flaring, green house and fluorinated gases, electronic wastes, hospital wastes, et cetera. These problems give rise to disturbing outcomes in the human society in the form of climate change, diseases, human casualties, environmental hazards and disasters, depletion of the ozone layer, loss of biodiversity and extinction of species, mass migrations, among others.

However, remarkable efforts have been made by environmentalists, scientists and cooperate bodies to address and resolve these environmental issues. As a result, lots of funds are employed into the waste management activities. Even though, some of these efforts have been rewarded with some outstanding results, male and female environmental officers seem to have no detailed assessment of the effectiveness of these activities in ameliorating the sufferings of community members. Male and female environmental officers have different perspectives about waste management activities and play distinct roles in protecting

the environment. Hence, women express stronger environmental concern, focus on daily actions like recycling and conservation, and are more attuned to the vulnerability of their communities to environmental harm. Men are more likely to engage in public, socio-political actions, though this doesn't diminish their concern. This shows that gender has no influence in waste management activities. Records in various local government areas showed that there are many completed and commissioned waste management activities in various communities for sustainable community development. There are also cases of uncompleted and/or jumbled waste management activities within the area that the communities are not interested in. From records available too, male and female environmental officers are sidelined in most of the waste management activities, mostly because they are not well exposed on what to do in most cases and how to carry the community along for effectiveness. Also, it seems that lots of investments put into these waste management activities are not commensurate with their outcome while some are outside the immediate needs of community members.

It is then necessary to determine the extent waste management activities are utilized in resolving environmental issues for Sustainable Community Development in South East States, of Nigeria so that waste management activities could be implemented better hopefully with all stakeholders involved. Therefore, the problem of the study posed as a question is:

Methodology

Research design: Descriptive Survey Design was used for the study.

Sample and sampling Technique: The sample size for this study was 420 respondents which comprised 297 male and 123 female members of Community Project

To what extent has waste management activities been utilized in resolving environmental issues for sustainable community development in South East States, Nigeria?

Purpose of the Study

The purpose of this study was to determine the extent to which waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria. Specifically, the study determined the extent to which:

1. waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria.

Research question.

1. to what extent are waste management activities utilized in resolving environmental issues for sustainable community development in South East States, Nigeria?

Hypothesis. There is no significant difference between the mean response of male and female members of community project management committee on the extent waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria.

Management Committee (CPMC) from the communities under study.

Instrument for data collection: a structured questionnaire was used to elicit information from the respondents.

Method of Data Analysis: Data was analysed using mean and standard deviation while hypothesis was tested at 0.05 level of significance and appropriate degree of freedom with t-test statistics.

Result

Research Question One: What is the extent to which waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria?

Table 1: Mean and Standard Deviation Ratings of the Respondents on the Utilization of Waste Management Activities in resolving Environmental Issues for Sustainable Community Development

S/N	Through waste management activities community members learn to;	Male N=297			Female N = 123			Overall N = 420		
		$\bar{x}$	SD	Decision	$\bar{x}$	SD	Decision	$\bar{x}$	SD	Decision
19	repair spoilt or broken items to reduce waste in the environment.	3.11	0.75	HE	2.01	0.81	LE	2.56	0.82	HE
20	subscribing to on-line newspapers to keep waste away from the environment	3.06	0.72	HE	2.10	0.83	LE	2.58	0.84	HE
21	purchase items that have enduring values to reduce wastes	2.07	0.81	HE	3.07	0.87	HE	2.57	0.85	HE
22	give out items they do not need like clothes, shoes, bags, et cetera to less privileged ones to reduce waste.	3.66	0.78	HE	3.06	0.72	HE	3.34	0.75	HE
23	observe the use of reduce, reuse and recycling method of waste management	3.60	0.87	HE	3.12	0.75	HE	3.36	0.76	HE
24	use household wastes to form organic manure for subsistence farming which keeps waste away from the environ.	3.40	0.76	HE	2.40	0.61	LE	2.90	0.60	HE
25	recycle item such as metal to reduce waste.	2.62	0.73	HE	3.22	0.71	HE	2.92	0.56	HE
26	purchase things in bulk to avoid single packaging.	3.66	0.78	HE	3.06	0.87	HE	3.34	0.76	HE
27	print on both sides of papers to minimize waste	3.20	0.72	HE	3.42	0.78	HE	3.31	0.77	HE
28	hire item like chair, which help to manage waste.	3.01	0.75	HE	2.95	0.72	HE	2.85	0.66	HE
Grand mean		3.14	0.78	HE	2.72	0.68	HE	2.97	0.73	HE

Key: N=Number of the respondents, HE=High Extent, $\bar{x}$ = Mean, SD= Standard deviation

Table 1 shows the results of the analysis conducted to determine the extent to which waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria. Table 3 also indicated that the mean ratings on the utilization of waste management activities in resolving environmental issues for sustainable development ranged from 2.57 to 3.36 with standard deviation values that ranged from 0.56 to 0.85. All the items had mean ratings above the 2.50 benchmark, implying that the members of community believed that waste management activities were utilized to a high extent in resolving environmental issues for sustainable community development in South East States, Nigeria. This is more evident in the grand mean of 2.97 which was also well above the 2.50 benchmark. The low standard deviation ratings on these items implies that the responses of the members of community on utilization of waste management activities in resolving environmental issues for sustainable community development were homogenous as their responses clustered fitly around the mean. The answer to research question one is that the waste management activities are utilized to a high

extent in resolving environmental issues for sustainable community development in South East States, Nigeria. These include: repairing spoilt or broken items to reduce waste in the environment, subscribing to on-line newspapers to keep waste away from the environment, purchasing items that have enduring values to reduce wastes, giving out items they do not need like clothes, shoes, bags, et cetera to less privileged ones to reduce waste, observing the use of reduce, reuse and recycling method of waste management. Others includes; using household wastes to form organic manure for subsistence farming which keeps waste away from the environ, recycling items such as: metals, plastic, glasses and polyethene reduces waste, purchasing things in bulk to avoid single packaging, printing on both sides of papers to minimize waste, and hiring items like chairs, tables, tent, coolers and ladders which help to manage waste. .

Hypothesis 1. There is significant difference between the mean responses of male and female members of community project management committee on the extent to which waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria

Table 2: Summary of the t-test Analysis of the mean Ratings of Male and Female Members of Community Projects Management Committee on the extent to which Waste Management Activities are utilized in Resolving Environmental Issues for Sustainable Community Development in South East States, Nigeria.

Gender	N	$\bar{x}$	SD	Df	t-cal	t-crit	Decision
Male	297	3.14	0.78	418	4.42	±1.96	S
Female	123	2.72	0.68				H ₀₃ Not accepted

Significant at $p < .05$, $df = 418$, critical t-value = ± 1.96

The t-test analysis in Table 2 above indicates that the t-calculated value is 4.42 while the critical t-value is ± 1.96 at .05 level of confidence. This implies that the calculated t-value is greater than the critical t-value. Thus, going by the decision rule, there is a significant difference in the mean scores of male and female members of community project management committee on the extent to which waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria.

Discussion of Findings.

The findings of the study on research question one showed that waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria. The findings showed that repairing spoilt or broken items to reduce waste in the environment, subscribing to on-line newspapers to keep waste away from the environment, purchasing items that have enduring values to reduce wastes, giving out items they do not need like clothes, shoes, bags, et cetera to less privileged ones to reduce waste, observing the use of reduce, reuse and recycling method of waste management, using household wastes to form organic manure for subsistence farming which keeps waste away from the environ, recycling items such as; metals, plastic, glasses and polythene reduces waste, purchasing things in bulk to avoid single packaging, printing on both sides of papers to minimize waste, and hiring items like chairs, tables, tent, coolers and ladders which help to manage waste are some of the ways in which waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria. The findings of

the study were in conformity with the view of Zafar (2020) who, while quoting the statement of National Environmental Standards Regulation Enforcement Agency opined that waste management is on how wastes generated by human activities are collected, transported, reduced, recycled, reused and disposed properly. This will help the members of the community to participate actively in the programme thereby, reducing the impacts and detriment of wastes on the health of man and his environment.

The findings of the study according to null hypotheses one showed that there was significant difference between the mean ratings of male and female members of community project management committee on the extent waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria. This implies that both male and female members of community project management committee shared different view on the extent waste management activities are utilized in resolving environmental issues in South East States, Nigeria.

Conclusion. The study therefore, concludes that waste management activities are utilized in resolving environmental issues for sustainable community development in South East States, Nigeria. As a result, government should continue to promote the activities of community project management committee in various communities for effectiveness and efficiency. They should also ensure that investments put into the sustainable community development activities are commensurate with their outcome and are not outside the immediate needs of the community. This will help the general public to understand, appreciate and promote the actions of community project management committee as activities that

would benefit them. Thus, the standard of living of the community members will be improved. It will also ensure sustainable community development in South East States, Nigeria.

Recommendations

1. The community project management committee should ensure proper waste management activities in the target communities as a method of resolving environmental issues in their various communities.
2. The members of community project management committee should ensure that the communities as the beneficiaries of their activities should be properly informed and have the direct impact of the activities.
3. The study recommended amongst others that the government should

encourage and support community project management committee into embarking on more sustainable community development activities for the good of the community.

4. Community project management committee members should ensure adequate sustainable community development activities in the target communities to resolve environmental issues and bring about sustainable community development in their various communities.
5. Members of the community should utilize the opportunities provided by the community project management committee members through development activities in their various communities to improve their standard of living.

REFERENCES

- Akinloye, A. O. (2018). Influence of waste management on environmental insecurity among residents in Lagos. *Journal of Research and Contemporary issues in Human Kinetics and Health Education*, 4(1), 22-31.
- Berg, C. (2017). *Shaping the future sustainably – types of barriers and tentative action principles: future scenarios of global cooperation—practices and challenges*. Käte Hamburger Kolleg.
- Budge, S., Argaw, A., Jamie, B., Brown, J., & Hutchings, P. (2022). Environmental sanitation and the evolution of water, sanitation and hygiene. *Bull World Health Organ.* 100(4), 286–288. doi: 10.2471/BLT.21.287137
- Cabaj, M. & Weaver, L. (2016). *Collective impact 3.0: An evolving framework for community change*. Canada: Tamarack Institute.
- Carrington, U. (2018). Assessment of community participation in community and social development projects (csdp) in Enugu State. *International Journal of Academic Management Science Research (IJAMSR)*, 7, (3), 30-37
- Harrington, L. M. B. (2016). Sustainability theory and conceptual considerations: a review of key ideas

- for sustainability, and the rural context. *Papers in applied geography*, 2, 365-382
- Hannah, T. (2018). *Give earth a chance, environmental activism in Michigan*. <http://michiganintheworld.history.lsa.umich.edu/environmentalism>
- Holly, E.R., & Mberu, B.U.(2015). Capitalizing on Nigeria's demographic dividend: reaping the benefits and diminishing the burdens. *African population studies*, 27, 2, 319-33
- National Policy on the Environment (2016). Ecolex | The gateway to environmental law. <https://www.ecolex.org> › details › legislation › national-...
- Okafor, S. N. (2019). *Influence of community education on participation of citizens in environmental protection activities in Anambra State, Nigeria*. Doreen Kobani.
- Okanlawon, S.A., Odunjo, O.O.,& Soyibo, O.O. (2024). *Challenges of Sustainable Environmental Management in Nigeria*. *Journal of economics and sustainable development*, 6, (5), 2222-2855
- Poulopoulos, S. G. (2016). *Environment and development. basic principles, human activities, and environmental implications*. Elsevier. <https://doi.org/10.1016/C2012-0-07133-5>
- Sagdiç, A. & Şahin, E. (2016). An assessment of Turkish elementary teachers in the context of education for sustainable development. *International Electronic Journal of Environmental Education*, 6 (2), 141-155.
- Torres, P. (2016). *Biodiversity loss: An existential risk comparable to climate change*. *The bulletin.org*. Taylor & Francis.
- Zafar, S. (2020). *Waste Management Challenges in Developing Nations. Powering a Greener Future*. BioEnergy Consult.